

PROTYAY DEY

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EDUCATION

University at Buffalo

PhD in Computer Science and Engineering, CGPA: 3.79 / 4.0

New York, US
Aug 2025 – Present

Indian Institute of Technology Ropar

MTech in Artificial Intelligence, CGPA: 9.09 / 10.0

Punjab, IN
Aug 2022 – Jun 2024

University of Engineering & Management, Jaipur

BTech in Computer Science and Engineering, CGPA: 9.27 / 10.0; Gold Medalist

Rajasthan, IN
Aug 2017 – Jun 2021

RESEARCH EXPERIENCE

Computer Vision and Pattern Recognition Lab, IIT Ropar

Junior Research Fellow – Federated Learning, Deep Learning

Punjab, IN
Apr 2025 – Jun 2025

- Developed FedRWA for disjoint-label federated learning, evaluated on **12 clients and 313 classes**.
- Achieved **18.6%** accuracy gain and **57.8%** lower client-wise variance across 3 backbones.

Image Processing, Security & Analytics Labs, IIT Ropar

Research Intern – Applied Deep Learning

Remote
Jul 2024 – Oct 2024

- Developed SPAIR-Swin with **20 high-entropy patches/image**, evaluated across **138 camera models**.
- Improved patch-level SCMI accuracy by up to **20.36%** across 4 forensic benchmarks.

mVizn Pte. Ltd.

Machine Learning Intern – 3D Computer Vision

Remote
Jan 2024 – Mar 2024

- Increased indoor point-cloud classification accuracy by **25%** using PointNet++.
- Improved IoU for underrepresented classes by **30%–45%** through class-imbalance handling.

SELECTED RESEARCH PROJECTS

MAGNET: Manifold-Aware Graph Diffusion for Connectome Generation

Mar 2026 – Present

- Proposed **MAGNET**, a manifold-aware graph diffusion model for valid brain connectome generation across **3 cohorts and 4,989 scans**.
- Enhanced class-conditional fidelity by up to **21%** and zero-shot subclass alignment by **33–57%**.

Execution-Aware Adaptation for SLM Multi-Agent Reasoning

Jan 2026 – Present

- Proposed **EAD**, a label-free runtime controller for SLM MAS reasoning across **6 benchmarks and 14 baselines**.
- Achieved **+7.40% overall gain** and **+12.85% average gain** on harder reasoning benchmarks.

Training-Free Quantization for Diffusion Transformers [\[GitHub\]](#)

Oct 2025 – Dec 2025

- Built a **4-stage TGQ pipeline** for training-free **W4A8** quantization using tensor-train informed grouping.
- Reduced **FID** by **1.2%** and **sFID** by **1.7%** on **10K** ImageNet-256 samples.

PUBLICATIONS

- P. Dey, A. S. Jadhav, K. Rana, R. Chakraborty, G. Sharma, P. Goyal, “Camera Model Identification with SPAIR-Swin and Entropy-Based Non-Homogeneous Patches,” [arXiv](#), 2025.
- K. Rana*, A. S. Jadhav*, **P. Dey***, V. Rathi, P. Goyal, G. Sharma, “SCMI30-IITRPR: Smartphone Camera Model Identification Dataset Comprising Similar and Random Content Image Sets,” [techRxiv](#), 2024.
- **P. Dey**, A. S. Jadhav, K. Rana, “An Effective CNN-Based Approach for Synthetic Face Image Detection in Pre-Social and Post-Social Media Context,” [CVIP](#), 2023.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, SQL, L^AT_EX, JavaScript, TypeScript

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib

Research Areas: Small Language Models, Multi-Agent Systems, Transformers, Diffusion Models, Generative AI

Tools: Git, GitHub, Linux, Jupyter Notebook, Google Colab, VS Code, Streamlit

Languages: Bengali Native, English Full Professional Proficiency, Hindi Full Professional Proficiency